

BADIRA —Clinical Definition of Preeclampsia

1. Preeclampsia: Core Clinical Concept

1.1 Preeclampsia in simple words

Preeclampsia is a pregnancy-related multisystem hypertensive disorder that usually develops after 20 weeks of pregnancy. It is more than “high blood pressure in pregnancy”: hypertension is accompanied by proteinuria and/or evidence of maternal organ involvement. It can affect multiple maternal organs and systems, including the kidneys, liver, blood, brain, lungs, and blood vessels, and can also affect the placenta and baby. Placental dysfunction and maternal vascular dysfunction are important parts of the disease, although preeclampsia is complex and multifactorial. A woman can have clinically important preeclampsia without obvious symptoms, which is why routine antenatal monitoring is important.

Key concept: High blood pressure is an important sign of preeclampsia, but it is not the whole disease.

1.2 When Can Preeclampsia Occur?

Preeclampsia typically develops after 20 weeks of pregnancy, and it is more common later in pregnancy. It can also develop after delivery, known as postpartum preeclampsia.

Preeclampsia occurring before 20 weeks is very rare, rather than impossible, and should prompt investigation for unusual associated conditions. Risk assessment can begin earlier in pregnancy, but it is important to distinguish risk of developing preeclampsia from an actual clinical diagnosis of preeclampsia.

1.3 Which Organs and Systems Can Be Affected?

Because preeclampsia is a multisystem disorder, its effects are not limited to blood pressure.

System	What Can Happen
Blood vessels	Dysfunction of the blood-vessel lining can contribute to increased vascular resistance and fluid leakage.
Kidneys	Kidney involvement can cause protein to leak into the urine and kidney function to decline.
Liver	Liver injury can cause increased liver enzymes and may be associated with upper-abdominal pain.
Blood	Platelet levels can fall, and severe disease such as HELLP syndrome can involve breakdown of red blood cells.

Brain	Neurological involvement can cause headache or visual disturbances and, in severe cases, seizures (eclampsia).
Lungs	Fluid can accumulate in the lungs (pulmonary edema), causing difficulty breathing.
Placenta / baby	Placental dysfunction can affect fetal growth and contribute to other pregnancy complications.

Important: These are possible effects of the disease. The specific organ findings used as clinical diagnostic criteria are covered separately in Section 2.3.

1.4 Why Preeclampsia Matters?

Preeclampsia matters because it can progress quickly and unpredictably, sometimes even when the woman initially feels well.

As the condition progresses, it can cause serious maternal complications, including eclampsia (seizures), stroke, HELLP syndrome, kidney or liver injury, pulmonary edema, and placental abruption.

It can also affect the pregnancy and baby through complications such as fetal growth restriction, medically indicated preterm delivery, and, in severe cases, fetal or maternal death.

Another important challenge is that management often involves timing. Definitive management ultimately requires delivery of the placenta. Other treatments, such as antihypertensive medication and magnesium sulfate, are used when appropriate to reduce complications and manage risk while clinicians determine the safest timing of delivery.

This can create a clinical balance between allowing more time for fetal maturity and avoiding increasing maternal or fetal risk if the condition worsens.

For this reason, early recognition, regular monitoring, and timely clinical assessment are important.

Preeclampsia also has implications beyond the pregnancy, as a history of the condition is associated with increased long-term cardiovascular risk.

2. Clinical Diagnosis of Preeclampsia

The diagnostic framework in this section primarily follows ACOG criteria. Other international guidelines, including NICE and ISSHP, differ in some details, so the guideline used should always be specified when defining labels for clinical data.

Preeclampsia is diagnosed using a combination of blood pressure measurements and evidence of proteinuria and/or maternal organ involvement. The basic clinical framework is: New-onset hypertension after 20 weeks + proteinuria OR qualifying maternal organ dysfunction. Therefore, hypertension is essential to the standard diagnostic framework, but proteinuria is not always required.

2.1 Blood Pressure Criteria

New-Onset Hypertension

New-onset hypertension means that blood pressure becomes abnormal after 20 weeks of pregnancy in a woman who was previously normotensive.

Systolic blood pressure ≥ 140 mmHg the upper number, representing arterial pressure when the heart contracts.

Diastolic blood pressure ≥ 90 mmHg the lower number, representing arterial pressure between heartbeats.

- Systolic ≥ 140 mmHg or diastolic ≥ 90 mmHg = hypertension.
- Systolic ≥ 160 mmHg or diastolic ≥ 110 mmHg = severe-range hypertension.

Either value crossing its threshold is sufficient; both do not need to be elevated.

Hypertension identified before 20 weeks is more consistent with chronic or pre-existing hypertension, although the woman's history and clinical context must be considered.

Confirmation of Hypertension

Ordinarily, an abnormal blood pressure is confirmed on two measurements at least 4 hours apart.

This is important because blood pressure can temporarily change because of factors such as anxiety, pain, physical activity, bladder fullness, incorrect cuff size, or white-coat effects.

Severe-Range Hypertension

Severe-range hypertension is:

- Systolic ≥ 160 mmHg, and/or
- Diastolic ≥ 110 mmHg

An important ACOG distinction: if a woman has gestational hypertension and develops severe-range blood pressure, she is diagnosed and managed as having preeclampsia with severe features, even without proteinuria or another severe feature.

Severe-range blood pressure is particularly important because very high blood pressure creates an immediate maternal risk, including increased risk of stroke. Unlike ordinary hypertension, it can be confirmed within a short interval so that treatment is not delayed.

Blood pressure is one of the most important routinely measurable signs used to identify and monitor hypertensive disorders during pregnancy. It can help detect abnormal changes even when a woman has no obvious symptoms. When blood pressure reaches the severe range, it is also a direct maternal danger, particularly because of the increased risk of complications such as stroke.

Because high blood pressure in pregnancy has several different causes, and they are not the same illness:

- A woman can have had high blood pressure before she was pregnant (chronic hypertension).
- She can develop high blood pressure during pregnancy with nothing else wrong (gestational hypertension).
- Or she can develop high blood pressure plus evidence that her organs are being affected — that is preeclampsia.

Two women can have identical blood pressure readings and have completely different conditions with completely different risks.

2.2 Proteinuria Criteria

Proteinuria means protein is present in the urine. In preeclampsia, it reflects kidney involvement, as damage to the kidney's filtering system allows protein to leak into the urine.

Method	Diagnostic Threshold
24-hour urine collection	≥300 mg per 24 hours
Protein-to-creatinine ratio (PCR)	≥0.3
Urine dipstick	2+ when quantitative methods are unavailable

2.3 Preeclampsia Without Proteinuria

A major point in the modern definition is that proteinuria is not required in every case of preeclampsia.

A woman with new-onset hypertension after 20 weeks can meet criteria for preeclampsia without proteinuria when qualifying new maternal organ dysfunction is present.

Qualifying findings include:

- Thrombocytopenia: platelet count $<100,000/\mu\text{L}$. Falling platelets indicate involvement of the blood/clotting system.
- Renal insufficiency: serum creatinine $>1.1\text{ mg/dL}$ or doubling of the woman's baseline in the absence of other renal disease. The "doubling of baseline" criterion is important because a clinically meaningful decline in kidney function may occur even while the absolute creatinine value remains below 1.1 mg/dL .
- Impaired liver function: liver transaminases $\geq 2\times$ the upper limit of normal and/or severe persistent right-upper-quadrant or epigastric pain that is not otherwise explained.
- Pulmonary edema: fluid accumulation in the lungs that can impair breathing and oxygenation.
- Neurological or visual features: new-onset headache that is unresponsive to medication and not explained by another diagnosis, or visual symptoms not explained by another diagnosis.

This is important because clinically significant preeclampsia can occur with little or no proteinuria.

2.4 The Diagnostic Logic

Step 1 — Is there new-onset hypertension after 20 weeks?

Step 2 — If yes, is proteinuria present?

Step 3 — If proteinuria is absent, is qualifying maternal organ dysfunction present?

If the required accompanying findings are present, the clinical framework for preeclampsia can be met.

Key principle: Preeclampsia is not defined by blood pressure alone, and it is not defined by proteinuria alone. The diagnosis depends on the combination of hypertension with the appropriate accompanying clinical findings.

3. Distinguishing Preeclampsia From Other Hypertensive Conditions

Condition	Key distinction
Chronic hypertension	Hypertension present before pregnancy or before 20 weeks
Gestational hypertension	New hypertension after 20 weeks without proteinuria or qualifying organ dysfunction

Preeclampsia	New hypertension after 20 weeks with proteinuria and/or qualifying organ dysfunction
Preeclampsia with severe features	Preeclampsia with severe-range BP or qualifying severe features

These disorders can share the same finding—high blood pressure—but they are not the same condition. Correct classification depends on when the hypertension began, the woman's previous history and baseline, and whether additional features develop.

Key concept: A single blood-pressure reading does not tell the whole story. Pregnancy timing, previous measurements, and new clinical changes help determine what that reading means.

3.1 Superimposed Preeclampsia

Superimposed preeclampsia occurs when a woman with chronic hypertension develops preeclampsia during pregnancy. It can be harder to recognize because her blood pressure may already be elevated and, in some cases, proteinuria may already exist.

Clinicians therefore need to consider change from the woman's own baseline, such as worsening hypertension or new clinical abnormalities.

This is an important example of why previous measurements matter: the meaning of a current value can depend on what was present earlier in pregnancy.

4. Symptoms, Warning Signs, and What Does NOT Diagnose Preeclampsia

Preeclampsia may produce symptoms, but symptoms are not a substitute for the clinical diagnostic criteria described in Section 2. Some women have no obvious symptoms, while many symptoms associated with preeclampsia can also occur for other reasons during pregnancy.

4.1 Symptoms That May Occur

Possible symptoms include:

- Persistent or severe headache, particularly when it does not improve with usual treatment.

- Visual disturbance, such as blurred vision, flashing lights, spots, or temporary changes in vision.

- Pain in the right upper abdomen or epigastric area.

- New or unusual nausea and vomiting later in pregnancy.

- Sudden or marked swelling, particularly of the face or hands.

- Breathlessness or difficulty breathing.

Reduced urine output.

Important: These symptoms should be taken seriously when clinically concerning, but none of them alone proves that a woman has preeclampsia.

4.2 Why Symptoms Alone Cannot Diagnose or Exclude Preeclampsia

1. Symptoms cannot exclude the disease. Preeclampsia can be present without obvious symptoms. A woman feeling well therefore does not rule it out.
2. Symptoms cannot diagnose the disease. Many possible symptoms are non-specific and can occur during pregnancy for reasons unrelated to preeclampsia.
3. Clinical context matters. Diagnosis depends on the defined clinical framework in Section 2. Even symptoms that appear within diagnostic criteria may require additional qualification and professional assessment

4.3 What Does NOT Diagnose Preeclampsia?

The following findings may be clinically useful or may prompt further assessment, but none should be treated as a diagnosis of preeclampsia by itself.

Finding	Why It Is Not a Diagnosis by Itself	What It Represents
One high blood-pressure reading	A single non-severe reading requires appropriate confirmation and interpretation rather than being treated as a diagnosis on its own.	A clinical measurement that may prompt repeat measurement or assessment
Headache	Headache is common and can have many causes. The relevant neurological criterion is more specific than simply reporting a headache.	A symptom requiring clinical interpretation
Swelling (oedema)	Swelling is common during pregnancy and is not, by itself, a diagnostic criterion for preeclampsia.	A non-specific finding
Nausea	Nausea can occur for many reasons during pregnancy and is not diagnostic on its own.	A non-specific symptom
A risk factor or family history	Risk factors change the probability of developing preeclampsia; they do not establish that the disease is currently present.	An input to risk assessment

An AI risk score	A risk score estimates probability; it does not satisfy the clinical diagnostic criteria for preeclampsia.	A prediction or risk estimate
First-trimester risk screening	Early screening estimates future risk and may guide prevention or surveillance; it is not a diagnosis of current preeclampsia.	Screening / risk estimation

Key principle: A symptom, risk factor, screening result, single measurement, or AI output can contribute useful information, but it must not be presented as equivalent to a clinical diagnosis.

Sources

1. American College of Obstetricians and Gynecologists (ACOG).
Gestational Hypertension and Preeclampsia. Practice Bulletin No. 222. Obstetrics & Gynecology. 2020.

2. National Institute for Health and Care Excellence (NICE).
Hypertension in pregnancy: diagnosis and management. NICE Guideline NG133.

[NICE NG133 — Hypertension in pregnancy](#)

3. International Society for the Study of Hypertension in Pregnancy (ISSHP).
Magee LA, Brown MA, Hall DR, et al. *The 2021 International Society for the Study of Hypertension in Pregnancy classification, diagnosis & management recommendations for international practice.* Pregnancy Hypertension. 2022;27:148–169.

[ISSHP 2021 Guideline](#)